Question ID: e1f9000d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is $3{,}364$ feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

Answer

A. 7.25

B. 14.50

C. 105.13

D. 210.25

Correct Answer: B

Rationale

Choice B is correct. It's given that the function h estimates that the object is $3{,}364$ feet above the ground when it's dropped at $t = 0$. Substituting $3{,}364$ for $h(t)$ and 0 for t in the function h yields $3{,}364 = -16(0)^2 + b$, or $3{,}364 = b$. Substituting $3{,}364$ for b in the function h yields $h(t) = -16t^2 + 3{,}364$. When the object hits the ground, its height will be 0 feet above the ground. Substituting 0 for $h(t)$ in $h(t) = -16t^2 + 3{,}364$ yields $0 = -16t^2 + 3{,}364$. Adding $16t^2$ to each side of this equation yields $16t^2 = 3{,}364$. Dividing each side of this equation by 16 yields $t^2 = 210.25$. Since the object will hit the ground at a positive number of seconds after it's dropped, the value of t can be found by taking the positive square root of each side of this equation, which yields $t = 14.50$. It follows that the function estimates the object will hit the ground approximately 14.50 seconds after being dropped.

Choice A is incorrect. The function estimates that 7.25 seconds after being dropped, the object's height will be $-16(7.25)^2 + 3{,}364$ feet, or $2{,}523$ feet, above the ground.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.